

Jude Crener Junior PIERRE

NLP & Machine Learning Researcher | Ph.D. Candidate in Information Science

Social Computing Laboratory, Nara Institute of Science and Technology (NAIST), Japan | Ikoma, Nara, Japan | +81 80 9756 3500 | judecrenerpierre02@gmail.com

PROFILE SUMMARY

NLP and machine learning researcher specializing in transformer-based models, biomedical information extraction, and LLM applications. Since 2022, I have designed, trained, and evaluated models for named entity recognition, word sense disambiguation, text classification, semantic similarity, and low-resource data augmentation using Python, PyTorch, Hugging Face Transformers, and BERT-based architectures. Additional experience includes fraud and operational risk analytics, REST API development, web development, and experimental model deployment.

TECHNICAL SKILLS

Programming: Python, C/C++, JavaScript, SQL

Machine Learning & Deep Learning: PyTorch, Keras, scikit-learn, Transformers, RNN/LSTM, model training, evaluation, hyperparameter experimentation

NLP & LLMs: Hugging Face Transformers, BERT, NER, information extraction, text classification, sentence embeddings, semantic similarity, LLMs, RAG, PEFT/LoRA, prompt engineering

Data Analysis: Pandas, NumPy, Matplotlib, time-series analysis, correlation analysis, data cleaning, preprocessing, annotation, quality control

Software & Deployment: Git/GitHub, HTML, CSS, Node.js, Flask, REST APIs, Hugging Face model deployment

Systems: MQTT, industrial HMI-to-cloud communication, AVEVA cloud systems, Linux

NLP / ML RESEARCH & DEVELOPMENT EXPERIENCE

Doctoral Researcher | Social Computing Laboratory, NAIST, Japan

Apr 2024 - Present (Expected Mar 2027)

- Conduct research on Transformer-based methods for biomedical information extraction, named entity recognition, word sense disambiguation, and low-resource learning.
- Design reproducible experimental workflows spanning text preprocessing, model training, hyperparameter experimentation, error analysis, and quantitative evaluation.
- Evaluate LLM-generated and biomedical text using BERTScore, sentence embeddings, lexical metrics, semantic-similarity analysis, and embedding-based visualization.
- Translate research questions into end-to-end NLP experiments using Python, PyTorch, Hugging Face, BERT-based architectures, and LLMs.

Graduate Researcher (Master) | Social Computing Laboratory, NAIST, Japan

Apr 2022 - Mar 2024

Research Assistant: May 2022 - Dec 2022

- Conducted medical NLP research to extract structured information from biomedical scientific literature.
- Applied named entity recognition to identify diseases and drugs across PubMed publications and analyzed longitudinal publication trends and disease-treatment relationships.
- Built Python workflows for text preprocessing, data cleaning, information extraction, statistical analysis, visualization, annotation, quality control, and model evaluation.

SELECTED NLP & AI PROJECTS

Persona-Driven Data Augmentation for Low-Resource Disease NER | Ph.D. Research

- Developed and evaluated a persona-driven LLM data augmentation framework for disease information extraction using GPT-based generation and BERT token classification; assessed text quality and semantic fidelity with BERTScore, BLEU, sentence embeddings, and semantic-similarity analyses. Published in JMIR Medical Informatics. DOI: [10.2196/87831](https://doi.org/10.2196/87831)

Medical Abbreviation Expansion & Word Sense Disambiguation | Ph.D. Research

- Developed and benchmarked approaches for ambiguous medical abbreviation expansion using minimal context, comparing BERT, Support Vector Machines, and interpretable statistical/context-based methods. First-author MEDINFO 2025 paper. DOI: [10.3233/SHTI251228](https://doi.org/10.3233/SHTI251228)

Medical Literature Trend & Relationship Analysis | M.Sc. Research

- Applied an NER model, time-series analysis, and correlation analysis to PubMed literature to investigate disease/drug trends and relationships. M.Sc. thesis.

LLM & NLP Applications | Personal Projects

- Built experimental applications with Qwen, Llama, and GPT-2, including RAG pipelines, PEFT/LoRA experiments, and open-source LLM/BERT applications for text generation, classification, and question answering; deployed NLP demos using Flask APIs and Hugging Face.

INDUSTRY EXPERIENCE

Communication Protocol Design Engineer, R&D Intern | Schneider Electric | Osaka, Japan Jan 2023 - Mar 2023

- Designed and implemented an MQTT-based communication protocol for industrial HMI systems using C/C++.
- Implemented communication functions enabling HMI devices to exchange data with cloud-based AVEVA systems, with attention to reliable data transfer and system constraints.

Business Risk Analyst | Digicel Group 2018 - 2021

- Analyzed transaction, customer activity, and billing data using Python, SQL, Pandas, Excel, and internal databases to identify anomalies, suspicious activity, fraud indicators, and potential revenue leakage.
- Investigated unusual usage patterns and data inconsistencies through reconciliation, validation, and analytical checks.
- Produced analytical reports and visualizations using Excel and Matplotlib to support operational risk assessment and decision-making.

Web Developer, Freelance | Remote 2017 - 2018

- Built frontend and backend applications using JavaScript, Node.js, Python, and Flask; developed and integrated REST APIs and third-party services.

ACADEMIC EXPERIENCE

Teaching Assistant | Nara Institute of Science and Technology (NAIST), Japan Oct 2023 - Jan 2024

- Supported graduate-level NLP coursework, coding activities, and student Q&A.

EDUCATION

Ph.D. Candidate in Information Science | NAIST, Japan Apr 2024 - Present (Expected Mar 2027)

Focus: Natural Language Processing, biomedical information extraction, and applied machine learning.

M.Sc. in Information Engineering | NAIST, Japan Apr 2022 - Mar 2024

Focus: Natural Language Processing, biomedical information extraction, and time-series analysis of biomedical literature.

B.Sc. in Computer Science, Minor in Economics | Ecole Supérieure d'Infotronique d'Haiti (ESIH) Sep 2014 - Jun 2018

PUBLICATIONS

Pierre JCJ, Nishiyama T, Peng S, Wakamiya S, Aramaki E. Persona-Driven Data Augmentation for Disease Name Recognition Across Rare and General Disease Corpora: A Comparative Evaluation. *JMIR Medical Informatics*. 2026. DOI: [10.2196/87831](https://doi.org/10.2196/87831)

Pierre JCJ, Andrade GHB, Peng S, Yada S, Wakamiya S, Aramaki E. Abbreviation Expansion using only a Single Word as Context. *In Proceedings of the 20th World Congress on Medical and Health Informatics (MEDINFO 2025)*, Taipei, Taiwan, 2025. DOI: [10.3233/SHTI251228](https://doi.org/10.3233/SHTI251228)

HONORS & LANGUAGES

MEXT Japanese Government Scholarship: 2021 - Present; selected for academic excellence.

Languages: French (Native) | English (Professional) | Japanese (Basic)